

AI Agent Evaluation — Final Metrics Table

Recommended 7-metric evaluation set for the multi-agent SDLC project. Ranges and thresholds are presented as DeepEval defaults where applicable; most agent-evaluation metrics do not have a universal research threshold.

Metric	Good Range	Bad Range	Threshold	Definition	Resource / Website
1. Plan Quality	≥ 0.5	< 0.5	0.5 — DeepEval current default	Measures whether the agent's plan is logical, complete, necessary, and suitable for achieving the task.	DeepEval: deepeval.com/docs/metrics-plan-quality Also: PlanBench — arxiv.org/abs/2206.10498
2. Plan Adherence	≥ 0.5	< 0.5	0.5 — DeepEval current default	Measures whether the agent actually follows the plan it generated during execution.	DeepEval: deepeval.com/docs/metrics-plan-adherence
3. Task Success Rate (TSR)	Higher is better; close to project target / human baseline	Low success rate	No universal threshold; define a project SLA or baseline	Percentage of evaluation tasks that are successfully completed according to a defined success check.	SWE-bench: swebench.com AgentBench: arxiv.org/abs/2308.03688 τ-bench: arxiv.org/abs/2406.12045
4. Trajectory / Process Evaluation	Higher quality / more correct trajectory	Incorrect, inconsistent, or poor execution path	No universal threshold; define task-specific criteria	Evaluates the sequence of agent decisions and actions, not only the final answer or outcome.	DeepEval agent metrics: deepeval.com/guides/guides-ai-agent-evaluation-metrics AgentBench: arxiv.org/abs/2308.03688
5. Step Efficiency	≥ 0.5; closer to 1 is better	< 0.5	0.5 — DeepEval current default	Measures whether the agent completes the task without unnecessary, repeated, speculative, or detour steps.	DeepEval: deepeval.com/docs/metrics-step-efficiency
6. Robustness & Recovery	Small performance drop + high recovery rate	Large performance drop + low recovery	No universal threshold; set maximum tolerated drop / minimum recovery	Measures how well the agent handles changed, noisy, adversarial, or failed conditions and recovers from errors.	AgentDojo: arxiv.org/abs/2406.13352 τ-bench: arxiv.org/abs/2406.12045
7. Pass@k	Higher is better; close to 1 is stronger	Low Pass@k	No universal threshold; choose k and target for the application	Probability that at least one of k generated attempts successfully solves a task. Especially useful for code generation.	HumanEval: arxiv.org/abs/2107.03374

Important Threshold Note

DeepEval current documentation: Plan Quality, Plan Adherence, and Step Efficiency use a default passing threshold of 0.5. Your earlier evaluation PDF used 0.7 as an example configuration for some stricter metrics; that is a project configuration, not a universal threshold. For TSR, Trajectory/Process, Robustness & Recovery, and Pass@k, the literature does not establish one universal good/bad cutoff. For your project, define targets using a baseline, human performance, risk level, or an agreed project SLA.

Recommended Interpretation for Your SDLC Agent

Reasoning: Plan Quality + Plan Adherence → checks whether the agent plans well and follows its plan.
Process: Trajectory/Process + Step Efficiency → checks whether execution is correct and efficient.
Outcome: Task Success Rate → checks whether the software-development task is actually completed.
Reliability: Robustness & Recovery + Pass@k → checks performance under failures/variation and repeated generation.

Primary resources: DeepEval Agent Metrics Guide: deepeval.com/guides/guides-ai-agent-evaluation-metrics; DeepEval Metrics Introduction: deepeval.com/docs/metrics-introduction; SWE-bench: swebench.com; AgentBench: arxiv.org/abs/2308.03688; AgentDojo: arxiv.org/abs/2406.13352; τ-bench: arxiv.org/abs/2406.12045; PlanBench: arxiv.org/abs/2206.10498; HumanEval: arxiv.org/abs/2107.03374.